TECHCRUSH CAPSTONE

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REGISTRATION NUMBER: AWS/2025/TC5/163

COURSE: CLOUD COMPUTING COHORT 5

GROUP: GROUP 2

PROJECT TITLE: CLOUD-BASED FILE STORAGE CLI

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Cloud-Based File Storage Capstone Report

1. Introduction

This project implements a **simple cloud-based file storage system** using **Azure Storage** and **Bash scripting**.
The system allows users to **upload, download, list, and delete files** from a container in Azure Storage using a command-line interface (CLI).
Deployment is **fully automated using GitHub Actions**, which sets up the storage account, container, and CLI environment with every push to the repository.

Objectives:

- Build a cloud storage system like Dropbox or Google Drive (CLI version)
- Automate deployment with Bash and GitHub Actions
- Implement logging to track all operations

2. Tools and Technologies

- **Azure CLI** – manage Azure Storage from the command line
- **Bash scripting** – CLI tool for file management
- **GitHub Actions** – automate deployment
- **Azure Storage Account and Blob Container** – cloud storage backend
- **Git / VS Code** – project management and coding environment

3. Deployment Steps

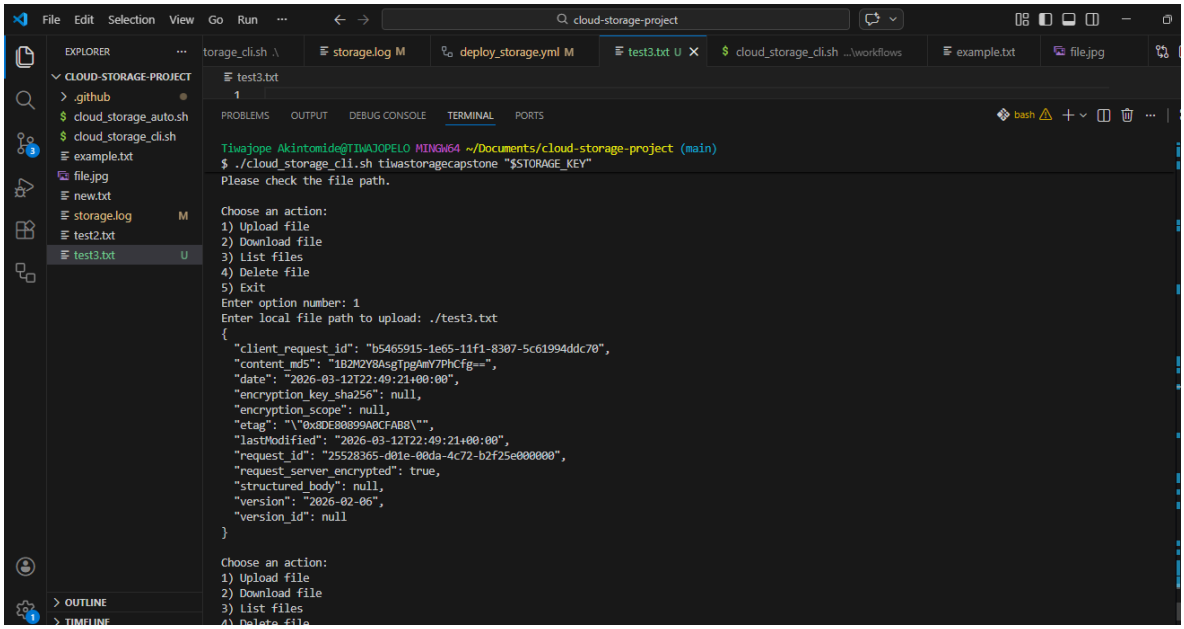
1. **Setup the GitHub Repository**
    - Added Bash scripts and GitHub Actions workflow .yml file.
    - Workflow automatically creates the storage account and container.
2. **Azure Setup**
    - Storage account created in resource group myResourceGroup.
    - Public blob access enabled.
3. **Running the CLI Script**
    - Execute the CLI using:

./cloud_storage_cli.sh <STORAGE_ACCOUNT_NAME> <STORAGE_KEY>

    - Menu prompts users to:
        1. Upload files
        2. Download files
        3. List files
        4. Delete files
        5. Exit

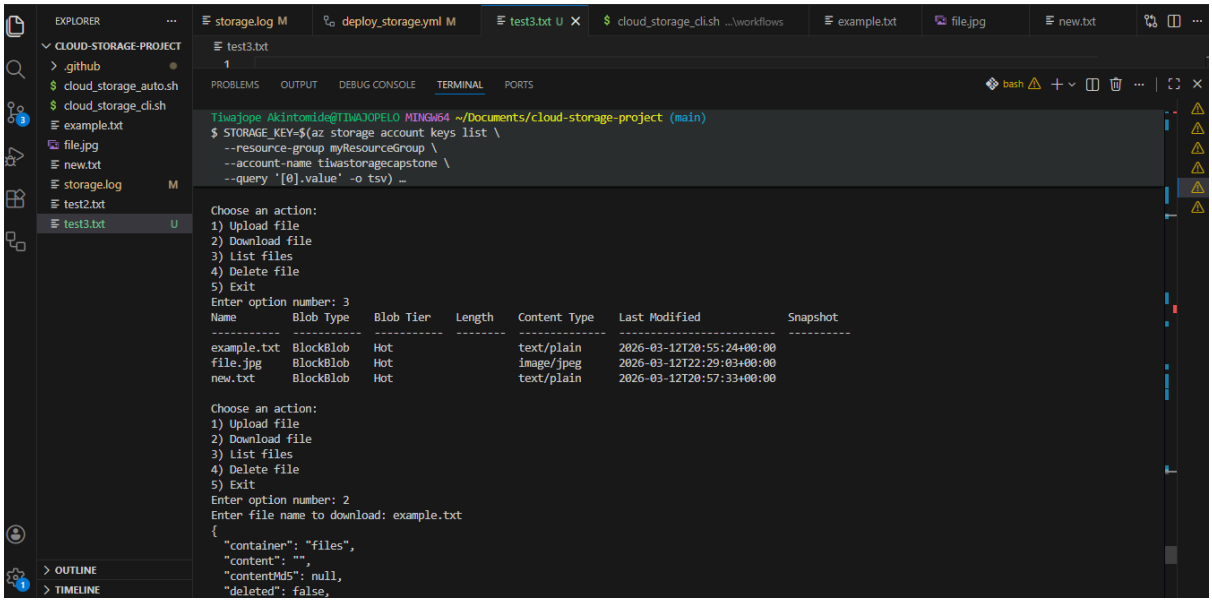
4. File Management Demonstration

a) Upload File



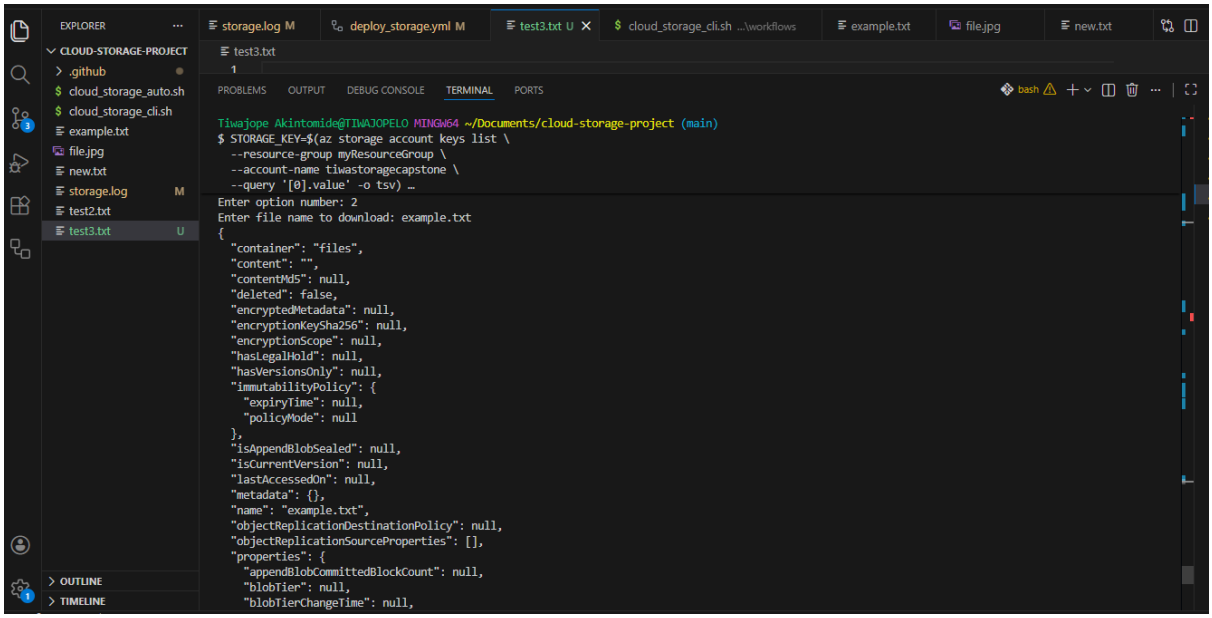
- File uploaded successfully to the Azure container.
- Log entry created in storage.log.

b) List Files



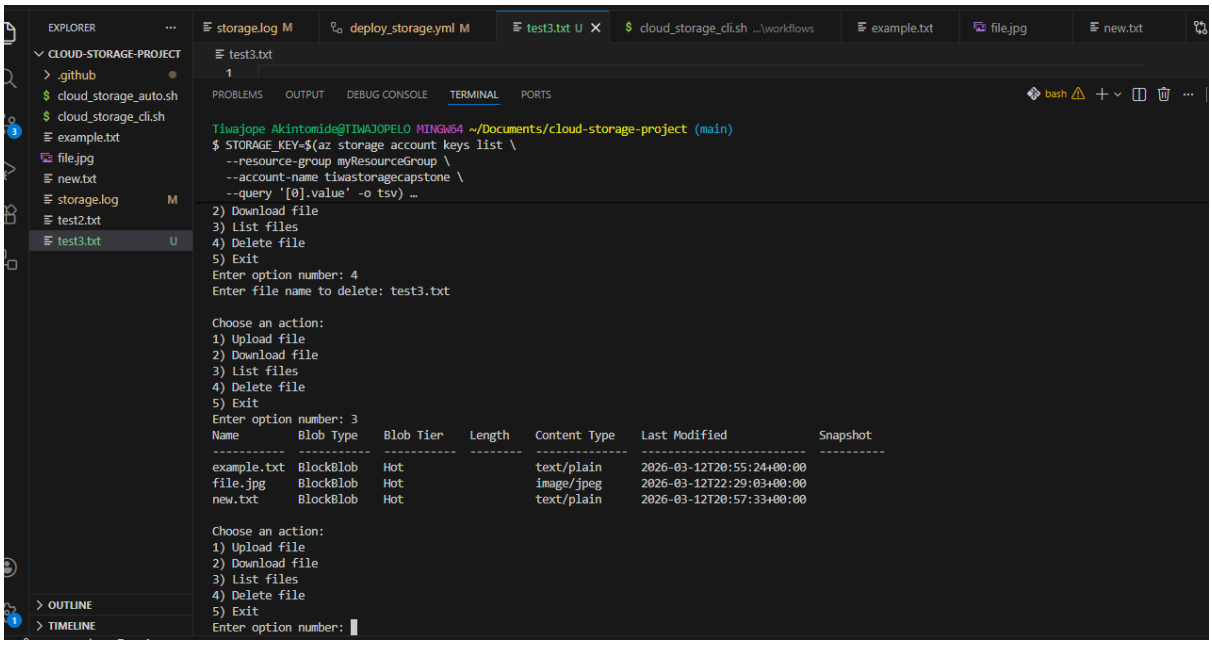
- Shows uploaded files in the container.

c) Download File



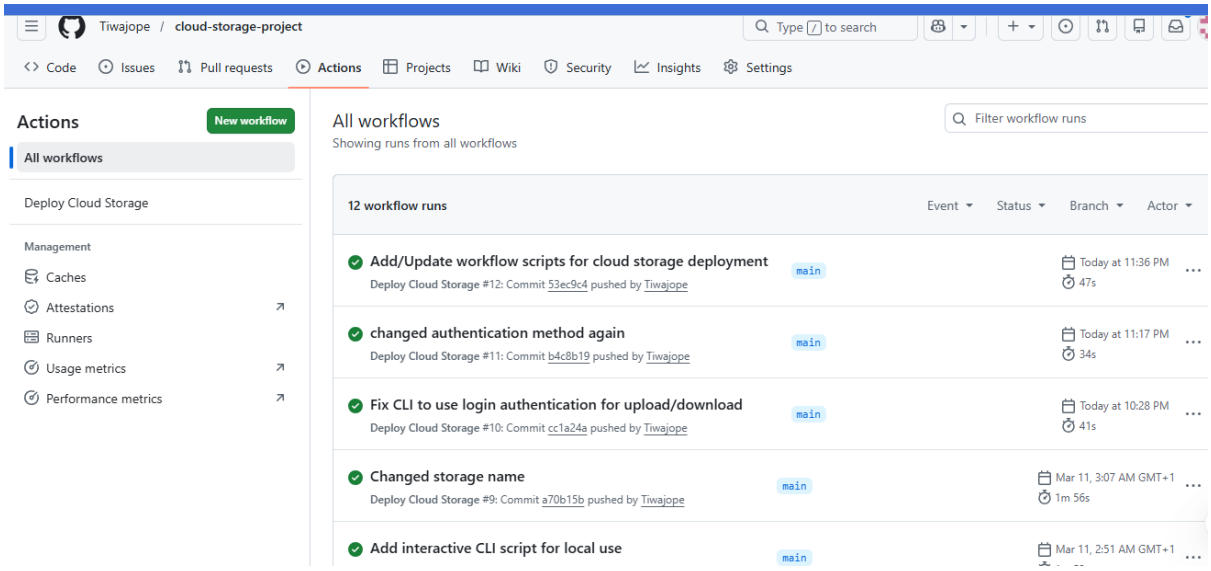
- File downloaded to local folder successfully.

d) Delete File



- File removed from container and verified in listing.

5. GitHub Actions Automation



- Shows workflow run green, indicating deployment succeeded automatically.
- Automates creation of storage account, container, and CLI setup.

6. Challenges and Solutions

1. **Authentication Errors**
    - Initially received “missing subscription” and “invalid key” errors.
    - Resolved by registering the Microsoft.Storage provider in Azure and using the correct storage account key.
2. **Container Not Created**
    - Fixed by ensuring proper Azure CLI commands and confirming resource group existence.
3. **DNS / Network Issues**
    - Solved by checking internet connectivity and Azure public endpoint access for the storage account.
4. **Container Appearing Empty**
    - When viewing the container in the Azure portal, no files were visible even though the container had been created.
    - This occurred because no files had yet been uploaded. A file was uploaded using the CLI script, after which the file appeared in the container.

Github link: https://github.com/Tiwajope/cloud-storage-project

7. Conclusion

The project successfully demonstrates a **cloud-based file storage system** using CLI, Azure Storage, and automated deployment.
All requirements were met: file upload, download, listing, deletion, logging, and automated deployment.